

A small offset, and why it matters

When we play the trained OP3 walking policy in MuJoCo and plot the commanded joint position against what the joint actually does, the two curves don't quite line up. The actual position lags the commanded position by a small, consistent amount, and its peaks are slightly smaller than the commands.

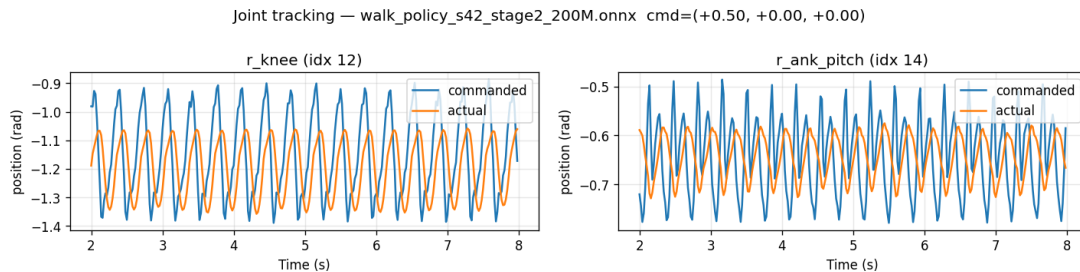


Figure 1: Knee and ankle-pitch tracking under a +0.5 m/s forward command, MuJoCo, seed-42 stage-2b. The actual position (orange) follows the commanded target (blue) but lagged by a small fixed amount.

That offset is *not* a tracking error. It is a property of the simulator: motors take a moment to move, joints have inertia, and the physics has its own natural delay and damping. The result is a predictable lag of a few tens of milliseconds between any commanded target and the joint reaching it.

The policy has learned the physics

The thing the policy is being rewarded for is *body* motion — moving forward at the right speed, turning at the right rate, not falling over. It is *not* being rewarded for matching a commanded joint exactly. So during training, it figured out which joint targets to issue *given the simulator's lag and damping*, so that the body ends up where it should be. The offset visible in the plot is the policy working with the physics rather than against it. In a real sense the inertia of the simulator is baked into what the policy commands.

You can read the offset as a signature of the actuator and physics the policy was trained on.

Why this is the sim-to-real gap

Two consequences follow:

- A different simulator (Webots) has a different lag and damping. The same commanded joint trajectory produces a different actual trajectory, and therefore a different body trajectory. Today we saw this in two flavours — wild jitter under Webots' built-in position controller, and a heavily-smoothed, low-amplitude walk under the ROBOTIS direct-control module's minimum-jerk filter.
- The real robot has its own lag and damping too — different motors, gears with backlash, finite controller bandwidth — and almost certainly slower than MuJoCo. So the same policy commands hit a different actuator chain and produce a different gait. That is what we observe deploying onto the OP3 hardware.

What the audit gives us

Matching the numerical gains, joint order, observation layout, and torque limits between training and deployment is necessary, and that is what most of our audit work has covered. But none of those checks catch a *bandwidth* mismatch. The phase-lag signature in the plot makes this visible: the audit confirms the actuator block, the plot confirms what the policy learned through that block, and the two together identify exactly where deployment platforms diverge.